54 - Spiral Matrix

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12

rows = 3
cols = 4

DIRECTION 4 States → ↓ ← ↑ loop

- On reaching the end
  - Change to the next direction
  - Mark the end (e.g col_end is the max j can be)

	Delta	Boundary Update
→	0, 1	col_end = j - 1
↓	1, 0	row_end = i - 1
←	0, -1	col_start = j + 1
↑	-1, 0	row_start = i + 1

directions = [(0,1), ...]

state = RIGHT

RIGHT, DOWN, ... = range(4)

while row_start <= row_end and col_start <= col_end

add val to res

check if move possible in curr direction

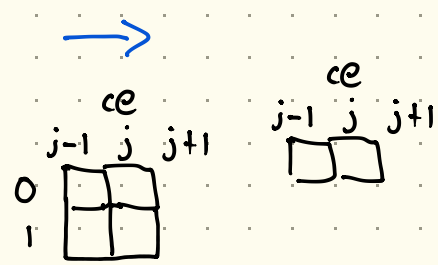
cant → update boundary (table)

rotate 90° (state + 1) % 4

move

if j + 1 > col_end:

col_end = j - 1



Edge Case: First traversal is two borders row_start ↓ col_end ←